JIAMING ZENG

jiaming@alumni.stanford.edu | Union City, CA | Google Scholar | Website | 901-338-4651

EDUCATION

Stanford University **Stanford, CA**
- M.S & Ph.D. in Management Science and Engineering *Sept. 2016 – June 2021*

Massachusetts Institute of Technology **Cambridge, MA**
- B.S. in Mathematics with Computer Science *Feb 2015*
- Minor in Applied International Studies/Concentration in German

WORK EXPERIENCE

Google, GCP Analytics AI **Sunnyvale, CA**
Senior Machine Learning Engineer / Tech Lead *June-2024 - Present*
- Driving e2e delivery and technical coordination of BigQuery Conversational Analytics from prototype to private preview in Nov 2025 and defining the North Star vision to guide future engineering scaling and development.
- Increased BigQuery user-acceptance rate by 13.3% and executability by 11.3% through self-consistency.
- Pioneered human-in-the-loop (HITL) work for disambiguation and explainability on agentic workflows.
- Lead Looker Query generation and delivering 27.8% accuracy and 7% executability improvements; owning service reliability, long-term strategy, and the migration to Java ADK.

AKASA **South San Francisco, CA**
Staff Machine Learning Researcher / AI Tech Lead *Sept. 2022 - June 2024*
- Lead R&D for human-in-the-loop AI systems for healthcare operations from inception to business impact.
- Researched techniques for extrapolating LLMs to long contexts on LongT5, MPT-7B, Llama2-7/13B, etc.
- Built LLM pipeline from infrastructure, data processing, continued pretraining, fine-tuning, evaluation, to launch.

X, The Moonshot Factory [Google X] **Mountain View, CA**
AI Residency *June 2019 – Sept. 2019*
- Developed technology to protect the ocean and improve sustainable fishing
- Delivered ML models to gain insights into fishing practices from limited data

NVIDIA, AI Infrastructure **Santa Clara, CA**
AI Research Intern *June 2018 – Sept. 2018*
- Optimized Bayesian neural networks to capture model uncertainty in active learning (NeurIPS 2018) (paper)
- Contributed Bayesian neural network training code to the official TensorFlow Probability repository

Oracle, Storage SW Quality Engineering **Burlington, MA**
Software Engineer *Mar. 2015 – Sept. 2016*
- Optimized machine learning models for the quality assurance process
- Ensured the quality and stability of the ZFS Storage Appliance device

NOAA Southwest Fisheries Science Center **San Diego, CA**
Ernest E. Hollings Scholar *May 2014 – Aug. 2014*
- Contributed to development of machine vision app in C# for fish detection and tracking in underwater videos
- Paper published in the IEEE Applications and Computer Vision Workshop 2015 (doi:10.1109/WACVW.2015.11)

BMW, R&D ConnectedDrive Group **Munich, Germany**
Software Engineer *May 2013 – Aug. 2013*
- Programmed a particle filtering system for the driver intent inference system to predict the driver's gaze direction
- Designed a calibration GUI for SmartTrack, a device used in data gathering for driver intent inference

RESEARCH EXPERIENCE

IBM Research, Computational Health Group **Cambridge, MA**
Postdoctoral Researcher *Sept. 2021 — Sept. 2022*
- Researched AI fairness and bias in healthcare treatments through analysis of clinical notes in MIMICIII (paper)
- Collaborated across industry and academia to research fairness through synthetic healthcare data (paper)
- Researched methods for search and retrieval on clinical documents with word embeddings and SBERT

Stanford Graduate School of Business, Athey Lab **Stanford, CA**
Advisor: Prof. Susan Athey *June 2017 – June 2021*
- Built interpretable AI tools to reduce selection bias on causal inference for cancer treatment studies
- Researched RL, Gaussian Process regression, and Bayesian optimization for parameter tuning on causal forests
- Simulated causal inference datasets and experiments to test various parameter tuning methods

Stanford, Management Science and Engineering, Decision Analysis Group **Stanford, CA**
Advisor: Prof. Ross Shachter *Sept. 2016 – June 2021*
- Modeled a radiologist's decision threshold for mammography screening with Bayesian networks
- Developed decision support tool for oncology decision making and treatment support
- Paper published to *Medical Decision Making* (doi: 10.1177/0272989X19832914)

Stanford, School of Medicine, Laboratory of Quantitative Imaging and Artificial Intelligence **Stanford, CA**
Advisor: Prof. Daniel Rubin *Sept. 2018 – June 2021*
- Trained NLP embedding models to identify and extract treatment information from EMR data (student abstract)
- Project funded by the Stanford Human-Centered Artificial Intelligence Institute Seed Grant (proposal)

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MIT Sloan School of Management, Prediction Analysis Lab — **Cambridge, MA**
Advisor: Prof. Cynthia Rudin — *Feb. 2014 – Feb. 2015*
- Developed interpretable models using machine learning algorithms to predict the probability of prisoner recidivism
- Mined the 1994 Prisoner Recidivism data for essential features that can be used in model development
- Paper published in the *Journal of Royal Statistical Society, Series A* (doi: 10.1111/rssa.12227)

MIT, Department of Mechanical Engineering — **Cambridge, MA**
Advisor: Prof. Alexie Kolpak — *Sept. 2012 – May 2013*
- Developed a theoretical "green" engine model that produces its own fuel by converting CO_2 into methanol
- Analyzed theoretical processes for heat transfer during carbon dioxide to methanol reaction

Tsinghua University, Department of Computer Science — **Beijing, China**
Advisor: Prof. Wang Xiaoge — *June 2012 – Aug. 2012*
- Optimized the performance of the Conjugate Gradient Solver in Los Alamos Lab's Parallel Ocean Program (POP)
- Operated experiments on supercomputers such as Tianhe1A and BlueLight through remote control

LEADERSHIP & TEACHING

Decision Analysis Working Group Seminar — **Stanford, CA**
Seminar Organizer — *Jan. 2020 – June 2021*

Women in Mathematics, Science, and Computational Engineering (WiMSCE) — **Stanford, CA**
Events Chair — *Sept. 2019 – Jun 2020*

MS&E 152/252: Introduction to Decision Analysis — **Stanford, CA**
Course Assistant for Prof. Ross Shachter — *Spring 2017, 2019; Winter 2021*

MS&E 120/220: Probabilistic Analysis — **Stanford, CA**
Course Assistant for Prof. Ross Shachter — *Fall 2018*

AWARDS, GRANTS, & SCHOLARSHIPS

- Stanford Institute of Human-Centered Artificial Intelligence Seed Grant Recipient (2019)
- 1st Place, USPROC Undergraduate Statistics Research Project Competition (2015)
- NOAA Ernest E. Hollings Scholarship (2013-2015)

PUBLICATIONS

- Bhanot, K., Baldini, I., Wei, D., **Zeng, J.**, & Bennett, K. (2023, August). Stress-Testing Bias Mitigation Algorithms to Understand Fairness Vulnerabilities. In Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society (pp. 764-774).
- **Zeng, J.**, Gensheimer, M. F., Rubin, D. L., Athey, S., & Shachter, R. D. (2022). Uncovering interpretable potential confounders in electronic medical records. Nature Communications, 13(1), 1014.
- Adam, H., Yang, M. Y., Cato, K., Baldini, I., Senteio, C., Celi, L. A., **Zeng J.**, Singh M & Ghassemi, M. (2022, July). Write it like you see it: Detectable differences in clinical notes by race lead to differential model recommendations. In Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society (pp. 7-21).
- **Zeng, J.**, Banerjee, I., Henry, A. S., Wood, D. J., Shachter, R. D., Gensheimer, M. F., & Rubin, D. L. (2021). Natural language processing to identify cancer treatments with electronic medical records. JCO Clinical Cancer Informatics, 5, 379-393.
- **Zeng, J.** (2020). Developing a Machine Learning Tool for Dynamic Cancer Treatment Strategies. *Proceedings of the AAAI Conference on Artificial Intelligence*, *34*(10), 13742-13743.
- **Zeng, J.,** Gimenez, F., Burnside, E. S., Rubin, D. L., & Shachter, R. (2019). A probabilistic model to support radiologists' classification decisions in mammography practice. *Medical Decision Making*, *39*(3), 208-216.
- **Zeng, J.,** Lesnikowski, A., & Alvarez, J. M. (2018). The Relevance of Bayesian Layer Positioning to Model Uncertainty in Deep Bayesian Active Learning. In *3rd Annual Bayesian Deep Learning Workshop, NeurIPS 2018*.
- **Zeng, J.,** Ustun, B., & Rudin, C. (2017). Interpretable classification models for recidivism prediction. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, *180*(3), 689-722.
- Cutter, G., Stierhoff, K., & **Zeng, J.** (2015). Automated detection of rockfish in unconstrained underwater videos using Haar cascades and a new image dataset: labeled fishes in the wild. In *Applications and Computer Vision Workshops (WACVW), 2015 IEEE Winter* (pp. 57-62). IEEE.
- [Non-academic] **Zeng, J.** (2019). My day as a double major. Firehose, *MIT Technology Review*. March 2019.

LANGUAGES

English (Native), Chinese (Native), German (Advanced), Python, PyTorch, R, Java, Golang, Kotlin, MATLAB, C#, LaTeX, bash